

Gebze Technical University

Computer Engineering Department

CSE 396 Computer Engineering Project

ASSIGNMENT 2

Module Interface & Header Design

Autonomous First Responder Fire & Rescue Robot

with Virtual Payload Delivery and Acoustic Localization

Submission Date: 30 March 2026

Instructor: Salih Sarp

Group: 7

Team Members & Assigned Modules

Name	Student ID	Assigned Module(s)
Nuri Ziya Kırtepe	210104004027	MOD-01, MOD-05
Ömer Nacar	210104004814	MOD-01, MOD-04
Gabil Rahimli	230104004902	MOD-01, MOD-02
Evrin Doğa Solmaz	230104004042	MOD-02, MOD-03, MOD-05
Dicle Çoban	220104004088	MOD-02, MOD-03, MOD-05
Uğur Anıl Güney	210104004011	MOD-02, MOD-03, MOD-04
Tuana Melisa Aksoy	230104004903	MOD-02, MOD-03, MOD-04
Fatma Öztürk	230104004152	MOD-02, MOD-04

1 Module Structure and Task Distribution (8-Person Team)

Person	Mod Emb HW	1: &	Mod 2: & Vision	AI	Mod 3: Au- dio & Nav	Mod Web STT	4: &	Mod Unity	5:
Gabil Rahimli	Secondary		Primary						
Ziya	Primary							Secondary	
Dicle Çoban			Secondary		Secondary			Primary	
Ömer	Secondary					Primary			
Uğur Anıl Güney			Secondary		Primary	Secondary			
Tuana Melisa Aksoy			Secondary		Secondary	Primary			
Evrin Doğa Solmaz			Primary		Secondary			Secondary	
Fatma Öztürk			Primary			Secondary			

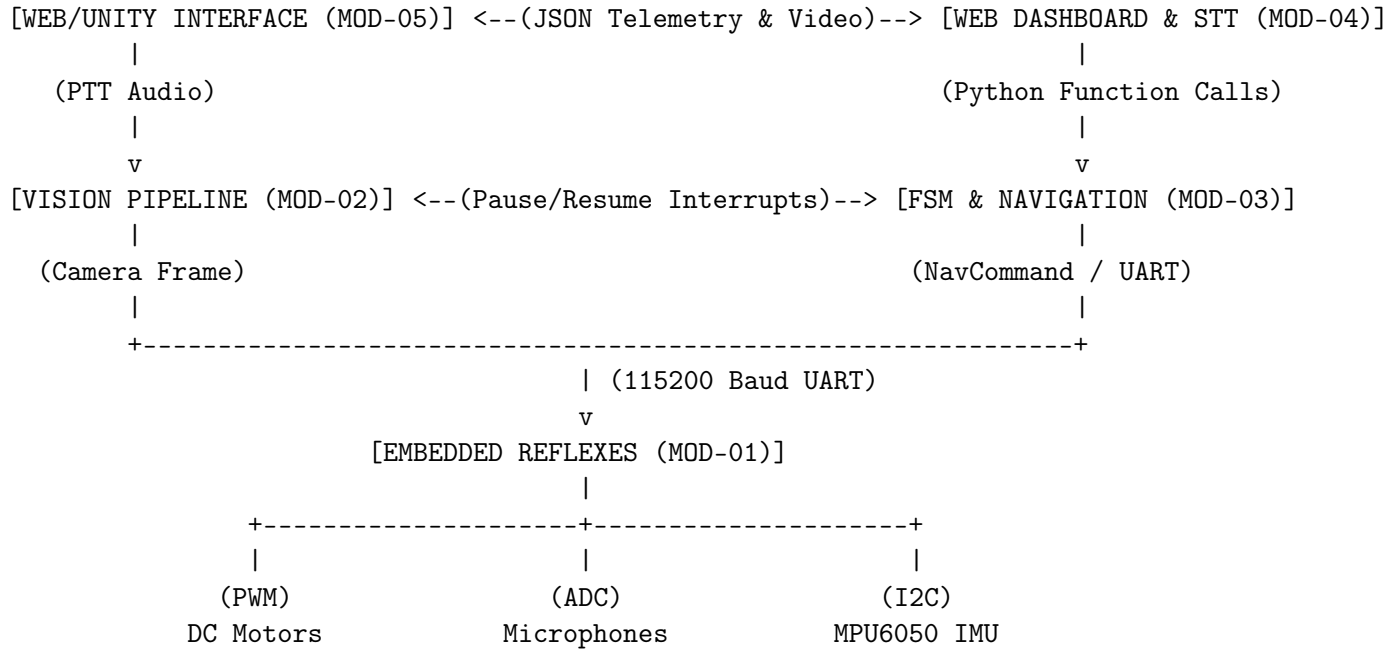
2 Module-Student Matching Table

Module ID	Module Name	Responsible Student(s)	Brief Role
MOD-01	Embedded & HW	Ziya, Ömer, Gabil	STM32 firmware: motor control, environmental sensing, stuck detection, power management, and UART bridge to Pi 5.
MOD-02	AI & Vision Pipeline	Fatma, Gabil, Evrim, Tuana, Uğur, Dicle	Camera handling, human detection, victim severity analysis, and runtime pause/resume support on Pi 5.
MOD-03	Audio & Navigation	Uğur, Evrim, Tuana, Dicle	IIR filtering, bearing computation, FSM acoustic branching, and Unity beam visualization calculations.
MOD-04	Web Dashboard & STT	Ömer, Tuana, Uğur, Fatma	Flask/WebSocket dashboard bridge, telemetry/video streaming, manual override routing, and offline STT on Pi 5.
MOD-05	Unity Digital Twin	Dicle, Ziya, Evrim	Operator dashboard: Map visualization, HUD, Push-to-Talk audio, and network client.

3 System / Module Overview

3.1 Overall Architecture

The project is a multi-module search and rescue robot system built on three main computational layers: an STM32 microcontroller for low-level hardware control, a Raspberry Pi 5 for high-level autonomy and AI inference, and a Unity-based operator dashboard.



3.2 Platform Summary

Category	Details
MCU	STM32F103C8T6 (Blue Pill)
SBC	Raspberry Pi 5 (8 GB RAM) + Official Active Cooler
Camera	Pi Camera Module V3
Microphones	3x MAX4466 electret microphone amplifier modules
Ultrasonic Sensors	HC-SR04 (front, back, left, right)
Sensors	MPU-6050 (IMU), DHT22 (Temperature), MQ-2 (Smoke)
Motor Driver	L298N H-Bridge
UART	115200 baud, 8N1
Web Dashboard	Flask + WebSocket based dashboard communication
STT Backend	Offline Vosk or Whisper.cpp style speech-to-text pipeline
Unity Side	Network client, HUD, map visualization, operator control panel

4 Inter-Module Communication Description

4.1 MOD-01 → Upper Layers: UART Telemetry

Field	From	To	Type	Dir.	Frequency
uart_telemetry_t	MOD-01	Pi-side FSM	UART string	01 → Upper	50 Hz (Every tick)

Representative wire format:

T:<temp>|S:<smoke>|ST:<stuck>|A_Ang:<acoustic_angle>|Y:<yaw>|UF:<front>|UB:<back>|...

4.2 MOD-02 ↔ MOD-04: Vision Interface Calls

Function	From	To	Type	Notes
get_latest_target()	MOD-04	MOD-02	Python Call	Returns TargetData or None.
pause_vision_pipeline()	MOD-04	MOD-02	Python Call	Called before STT to free RAM.
resume_vision_pipeline()	MOD-04	MOD-02	Python Call	Called after STT concludes.

Target Data schema:

TargetData(pos_x, pos_y, distance_cm, severity, confidence)

4.3 MOD-04 ↔ MOD-05: Dashboard Telemetry & Commands

Function / Payload	From	To	Type	Notes
broadcast_telemetry()	MOD-04	MOD-05	WebSocket	Sends AugmentedStatusReport.
stream_video_frame()	MOD-04	MOD-05	WebSocket	Streams compressed JPEG frames.
Operator Command	MOD-05	MOD-04	JSON	Receives manual override payload.
process_audio_blob()	MOD-05	MOD-04	Byte[]	Converts incoming .wav bytes.

Augmented Status Report Schema (JSON):

```
AugmentedStatusReport(  
    pos_x=12.5, pos_y=8.0,  
    temperature=25.1, smoke_detected=False,  
    victim_status="TRAPPED", is_stuck=False,  
    priority_level=1, acoustic_hit=True, acoustic_angle=-45.0  
)
```

Voice Command Schema (Processed internally in MOD-04):

```
VoiceCommandData(  
    raw_text="return home", intent="RETURN_HOME", confidence=0.91  
)
```

5 Header File Summary

Header / Interface File	Module	Lang	Key Types Exposed
uart_comm.h	MOD-01	C	uart_direction_t, uart_telemetry_t
environment_sensors.h	MOD-01	C	env_sensor_type_t, environment_data_t
motor_control.h	MOD-01	C	motor_status_t, motor_direction_t
stuck_detection.h	MOD-01	C	stuck_status_t, stuck_detection_result_t
pwr_management.h	MOD-01	C	pwr_status_t
ai_vision_interface.py	MOD-02	Py	TargetData, IVisionPipeline
camera_internal.py	MOD-02	Py	CameraFrame, CameraInternal
human_detector_internal.py	MOD-02	Py	CameraBBBox, HumanDetection
victim_analyzer_internal.py	MOD-02	Py	VictimAnalyzerInternal
fsm_acoustic.h	MOD-03	C	fsm_state_t, fsm_acoustic_result_t
acoustics_iir.h	MOD-03	C	acoustics_result_t, acoustics_nav_cmd_t
acoustic_homing.py	MOD-03	Py	AcousticTelemetry, NavCommand
MapManager_AcousticBeam.cs	MOD-03/05	C#	AcousticBeamData, AcousticBeamStyle
comms_dashboard_interface.py	MOD-04	Py	AugmentedStatusReport, IWebDashboard
stt_engine_interface.py	MOD-04	Py	VoiceCommandData, ISTTEngine
DataContracts.cs	MOD-05	C#	VictimStatus, TelemetryData
INetworkClient.cs	MOD-05	C#	INetworkClient interface
AudioManager.cs	MOD-05	C#	AudioCaptureState, IAudioManager
MapManager.cs	MOD-05	C#	MapManagerConstants, MapManager
UIManager.cs	MOD-05	C#	UIManagerConstants, UIManager

6 Integration Change Log

Code	File / Change	Reason
R1	uart_comm.h: Defined UART telemetry & cmd interface.	Base embedded communication contract.
R2	environment_sensors.h: DHT22 and MQ-2 abstraction.	Separates sensing from other MCU tasks.
R3	motor_control.h: L298N interface and PWM API.	Encapsulates robot motion control.
R4	stuck_detection.h: IMU stuck detection contract.	Supports autonomous recovery behavior.
R5	ai_vision_interface.py: Public MOD-02 interface.	External access to target, pause/resume.
R6	camera_internal.py: Camera and FPS management.	Separates hardware control from AI.
R7	human_detector_internal.py: Detection helpers.	Modularizes YOLO detection logic.
R8	victim_analyzer_internal.py: Severity helper.	Separates VLM analysis from detection.
R9	fsm_acoustic.h: FSM acoustic behavior interface.	Transitions for acoustic navigation logic.
R10	acoustic_homing.py: Telemetry/nav Python bridge.	Connects acoustics to higher-level FSM.
R11	comms_dashboard_interface.py: Flask/WebSocket API.	Matches MOD-04 comms role in root.
R12	stt_engine_interface.py: Offline STT contract.	Matches MOD-04 speech-processing role.
R13	UIManager.cs, MapManager.cs: Operator HUD layers.	Completes Unity-side monitoring.

7 Known Risks and Open Issues

- **OOM Risks During STT Integration:** Running YOLO, VLM, and Vosk/Whisper simultaneously on the Pi 5 will severely stress RAM. While `pause_vision_pipeline()` is defined, transition latency requires empirical benchmarking.
- **Mechanical Trim Calibration:** Lacking wheel encoders, mapping relies on dead-reckoning. Differences in DC motors may cause drift. The `motor_set_trim()` constants are pending arena tests.
- **Unity Coordinate Scaling:** The exact scaling factor between the Pi 5's Grid and the Unity World-Space (`GridToWorldPosition`) is unconfirmed.
- **Python Interfaces vs. Headers:** MOD-02 and MOD-04 expose Python ABC interfaces instead of C headers due to the framework requirements (Flask, PyTorch).
- **Placeholder IDs:** Some student IDs remain placeholders in module-level documentation and need finalization before presentation.
- **WebSocket Constants:** Exact WebSocket event names in MOD-04 are still marked as TODO in the repository README.

8 Conclusion

This report formally summarizes the module interfaces, public headers, and communication contracts of the Group 7 Search & Rescue Robot project. The resulting design robustly captures the division of embedded (MOD-01), AI (MOD-02), acoustic (MOD-03), dashboard/STT

(MOD-04), and Unity (MOD-05) responsibilities, satisfying all requirements of Assignment 2 and preventing integration bottlenecks in subsequent phases.